ASR-Style Transcript Normalization Removes Useful Pragmatic Signal for Dialogue Act Classification

Lucky Onyekwelu-Udoka | Md Shafiqul Islam

IOWA STATE UNIVERSITY
COLLEGE OF ENGINEERING
ELECTRICAL & COMPUTER ENGINEERING

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KEY FINDINGS: **Heavy normalization hurt every neural model on every corpus across all five seeds.** But filler-only effects changed with the corpus and lexicon, revealing that what is deleted matters.

THE QUESTION

Is transcript cleanup harmless?

ASR pipelines often remove pauses and punctuation or expand colloquial forms. Those choices can erase hesitation, turn-management, and stance cues before an NLP model ever sees them.

REFERENCE **"well, uh... I think so. uh-huh."**

CLEANED **"well i think so"**

The words remain. The interactional signal does not.

CONTROLLED EXPERIMENT

Three conversational registers

SWITCHBOARD	MRDA	SLUE-HVB
Telephone	Meetings	Call-center
27 labels | 34.6k / 9.1k	12 labels | 75.1k / 16.7k	16 labels | 11.3k / 6.1k

9 CONDITIONS	6 MODELS	5 SEEDS	PAIRED TESTS
gold, fillers, punctuation, heavy...	TF-IDF, BiLSTM, 4 transformers	key conditions across corpora	cluster bootstrap + McNemar

Nine transformations, applied to both train and test

KEEP	gold text	FILL	remove fillers	PUNCT	remove punctuation
COLLOQ	expand forms	DISC	remove markers	ALPHA	alphanumeric
LIGHT	light clean	HEAVY	aggressive clean	NOISE	synthetic ASR

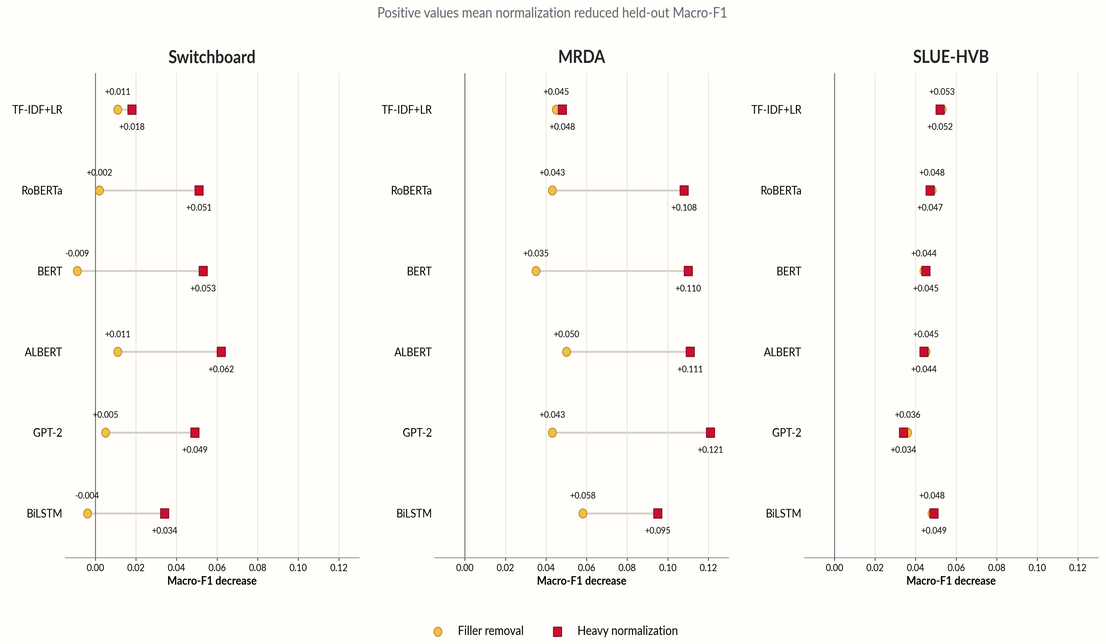
WHY THIS DESIGN MATTERS

- Separates deliberate normalization from recognition error.
- Uses transparent TF-IDF plus contextual neural models.
- Checks seeds, clusters, lexicon choice, and real Whisper output.
- Reports Macro-F1 so low-frequency dialogue acts still matter.

Probe question: does deleting a cue change pragmatic inference?

MAIN EVIDENCE

Aggressive cleanup is consistently harmful; filler-only removal is context-dependent



Cardinal squares (heavy normalization) stay to the harmful side. Gold circles (filler removal) cross zero on Switchboard, so the effect is not universal.

75 / 75
seed x model x corpus comparisons harmed by heavy normalization

50 / 50
neural filler-removal comparisons harmed on MRDA + HVB

21 / 36
paired McNemar tests survive Bonferroni correction

WHAT BREAKS?

SWB BACKCHANNEL	MRDA FLOOR HOLDER	HVB BACKCHANNEL
F1 -0.068	Recall 0.495 -> 0.330	F1 0.480 -> 0.163
recall/precision loss	turn-management cue	class token removed

MECHANISM CHECKS

Punctuation is signal

Punctuation-aware TF-IDF exposes losses hidden by the default tokenizer.



Lexicon choice changes the story

Original lexicon

um uh erm mm hmm mhm uh-huh huh

Restricted control: um uh erm

MRDA	SLUE-HVB
harm persists	effect vanishes
Delta 0.015-0.039	Delta -0.006 to 0.001

Why? The original inventory emptied 38 of 50 HVB backchannel utterances.

REAL-ASR STRESS TEST

Whisper-base.en HVB 18.3% WER | MRDA 41.0% WER

- MRDA: all six models degrade (Delta 0.137-0.282).
- HVB: matched ASR train/test; five models decline, GPT-2 slightly improves.
- Protocols differ, so effect sizes are not directly comparable.

DECISION RULE

Do not normalize by default.

Treat fillers, punctuation, and backchannels as features. Ablate each operation on the downstream task before deployment.

CODE, DATA & FULL RESULTS

SCAN

github.com/luckymaduabuchi/asr-normalization-dialogue-acts

Scripts, per-seed predictions, confidence intervals, and condition-level outputs.

TAKEAWAY **Transcript normalization is a modeling decision, not harmless cleanup.**

Key references: Clark & Fox Tree (2002); Core & Allen (1997); Stolcke et al. (2000); Zelasko et al. (2021); Kubis et al. (2023); Radford et al. (2023).

lucky@iastate.edu | shafiqul@iastate.edu